STAT 2020 Discussion Slides

Week 7: Poisson, Integration, and Continuous Distributions

Shuo Li

University of Iowa

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2. Poisson Rate and Window Size
3. Integration Methods (Quick Review)
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Agenda

- Poisson means under different time intervals and regions
- Quick review of integration methods
- One continuous piecewise example:
  - CDF from PDF
  - PDF from CDF
  - $\mathbb{E}(X)$ and $\mathbb{V}(X)$
- Extra practice problems (quick drills)
- Reminder on exam-question policy
- Attendance check (mental math)

Poisson Distribution Formula

If $X \sim \text{Pois}(\mu)$, then

$$P(X = k) = \frac{e^{-\mu}\mu^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

In rate form:

$$X = N_t \sim \text{Pois}(\lambda t) \quad \Longrightarrow \quad P(N_t = k) = \frac{e^{-\lambda t}(\lambda t)^k}{k!}.$$

Poisson: Time-Interval Example

Assume arrivals follow $N_t \sim \mathrm{Pois}(\lambda t)$.

If average arrivals are 3 per 10 minutes, then

$$\lambda = 0.3 \text{ calls/minute.}$$

Expected counts scale linearly:

$$\mathbb{E}(N_5) = 1.5, \quad \mathbb{E}(N_{20}) = 6, \quad \mathbb{E}(N_{60}) = 18.$$

How Interval Length Changes the Mean

If rate is λ per minute, then for any interval length t:

$$N_t \sim \mathrm{Pois}(\lambda t), \qquad \mu_t = \mathbb{E}(N_t) = \lambda t.$$

So when the interval is multiplied by c, the mean is also multiplied by c.

$$t \to ct \quad \Longrightarrow \quad \mu_t \to c\mu_t.$$

Example with $\lambda = 0.3$:

$$\mathbb{E}(N_{10}) = 3,\ \mathbb{E}(N_{20}) = 6,\ \mathbb{E}(N_{30}) = 9.$$

Poisson: Region-Size Example

Suppose event intensity is $\rho = 2$ per square mile per day.

For area A,

$$N_A \sim \text{Pois}(\rho A), \quad \mathbb{E}(N_A) = \rho A.$$

Examples:

$$A = 1 \Rightarrow \mathbb{E}(N_A) = 2, \quad A = 3.5 \Rightarrow \mathbb{E}(N_A) = 7, \quad A = 10 \Rightarrow \mathbb{E}(N_A) = 20.$$

Integration Methods: Quick Review

- Substitution
- Integration by parts
- Partial fractions
- Symmetry tricks
- Piecewise integration
- Optional advanced method

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du, \qquad \int u\,dv = uv - \int v\,du.$$

Common Integral Forms to Recognize

$$(1) \quad \int 1\,dx = x + c$$

$$(2) \quad \int m\,dx = mx + c$$

$$(3) \quad \int x^n\,dx = \frac{x^{n+1}}{n+1} + c$$

$$(4) \quad \int \sin x\,dx = -\cos x + c$$

$$(5) \quad \int \cos x\,dx = \sin x + c$$

Common Integral Forms to Recognize (Cont.)

(6) $$\int \sec^2 x \, dx = \tan x + c$$

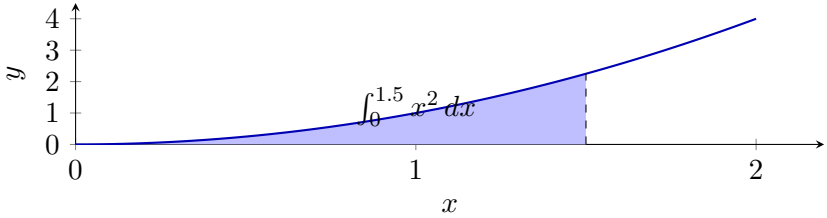
(7) $$\int \frac{1}{x} \, dx = \ln x + c$$

(8) $$\int e^x \, dx = e^x + c$$

(9) $$\int a^x \, dx = \frac{a^x}{\ln a} + c$$

What Integration Means (Area View)

$\int_a^b f(x)\,dx$ is the signed area between $y = f(x)$ and the x-axis.



Continuous Piecewise PDF

Define

$$f_X(x) = \begin{cases} x, & 0 < x \leq 1, \\ 2 - x, & 1 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

This is a continuous distribution, so $f(X = x) = 0$ for every fixed x.

Derive CDF from PDF

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ \dfrac{x^2}{2}, & 0 < x \leq 1, \\ 1 - \dfrac{(2-x)^2}{2}, & 1 < x < 2, \\ 1, & x \geq 2. \end{cases}$$

Recover PDF from CDF

Differentiate piecewise:

$$f_X(x) = F_X'(x) = \begin{cases} x, & 0 < x < 1, \\ 2 - x, & 1 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Point values (for example at $x = 1$) do not change probabilities.

Compute $\mathbb{E}(X)$ and $\mathbb{V}(X)$

$$\mathbb{E}(X) = \int_0^1 x \cdot x \, dx + \int_1^2 x(2-x) \, dx = 1.$$

$$\mathbb{E}(X^2) = \int_0^1 x^2 \cdot x \, dx + \int_1^2 x^2(2-x) \, dx = \frac{7}{6}.$$

$$\mathbb{V}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 = \frac{7}{6} - 1 = \frac{1}{6}.$$

Practice 1: Poisson Time Scaling

A store gets on average 4 customers every 20 minutes. Assume Poisson arrivals.

- Find λ (per minute)
- Find $\mathbb{E}(N_{30})$

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- Find $\mathbb{E}(N_{30})$

Answer:

$$\lambda = \frac{4}{20} = 0.2, \qquad \mathbb{E}(N_{30}) = 0.2 \times 30 = 6.$$

Practice 2: Poisson Region Scaling

Event intensity is $\rho = 1.5$ per square mile per day. For area $A = 8$, let $N_A \sim \text{Pois}(\rho A)$.

- What is $\mathbb{E}(N_A)$?

Practice 2: Poisson Region Scaling

Event intensity is $\rho = 1.5$ per square mile per day. For area $A = 8$, let $N_A \sim \text{Pois}(\rho A)$.

- What is $\mathbb{E}(N_A)$?

Answer:

$$\mathbb{E}(N_A) = \rho A = 1.5 \times 8 = 12.$$

Practice 3: Which Method?

Match each integral with a method:

$$\text{(a)} \int 2x\cos(x^2)\,dx, \qquad \text{(b)} \int xe^x\,dx, \qquad \text{(c)} \int \frac{1}{x^2-1}\,dx.$$

Practice 3: Which Method?

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$$\text{(a)} \int 2x\cos(x^2)\,dx, \qquad \text{(b)} \int xe^x\,dx, \qquad \text{(c)} \int \frac{1}{x^2-1}\,dx.$$

Answer:

- (a) substitution
- (b) integration by parts
- (c) partial fractions

Practice 4: Quick CDF Value

Let

$$f_X(x) = \begin{cases} x, & 0 < x \leq 1, \\ 2 - x, & 1 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Compute $P(X \leq 1)$.

Practice 4: Quick CDF Value

Let

$$f_X(x) = \begin{cases} x, & 0 < x \leq 1, \\ 2 - x, & 1 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Compute $P(X \leq 1)$. **Answer:**

$$P(X \leq 1) = F_X(1) = \frac{1^2}{2} = \frac{1}{2}.$$

Practice 5: Probability by Area

Using the same $f_X(x)$, compute $P(1 < X < 2)$.

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Using the same $f_X(x)$, compute $P(1 < X < 2)$. **Answer:**

$$P(1 < X < 2) = 1 - P(X \leq 1) = 1 - \frac{1}{2} = \frac{1}{2}.$$

Class Reminder: Exam Questions

- Specific exam questions are not discussed during class sessions.
- Exam-related questions can be addressed in office hours.

Take-Home Messages

- In Poisson models, means scale linearly with interval/region size.
- Piecewise integration gives a clean route to CDF/PDF/$\mathbb{E}$/$\mathbb{V}$.
- Keep exam-specific conversations in office hours.

In-Class Attendance Check (Quick)

Suppose calls follow a Poisson model. If average calls are 2 per 10 minutes, what is

$$\mathbb{E}(N_{30})?$$

Please submit: Full Name; Section Number; Your Answer

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Please submit: Full Name; Section Number; Your Answer

Answer:

$$\mathbb{E}(N_{30}) = 6.$$